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From screens to serenity: evaluating the effect of digital detox on mental and physiological health

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Abstract

Background Excessive screen time is associated with psychological distress and physiological dysregulation. We evaluated a 2-week digital detox in medical students, focusing on key outcomes: cortisol, C-reactive protein (CRP), interleukin-6 (IL-6), heart-rate variability (HRV), and psychometrics (Perceived Stress Scale [PSS], Generalized Anxiety Disorder-7 [GAD-7]).

Methods In a randomized three-group trial at two medical colleges, participants were allocated 1:1:1 to (i) detox + alternative activities, (ii) screen-time reduction only, or (iii) control. Randomization used stratified block procedures (campus × academic year × gender). We employed a convergent parallel mixed-methods approach, collecting qualitative feedback alongside quantitative measures, and integrating at interpretation. Compliance was tracked via device-level analytics and daily logs. Pre- and post-assessments included HRV, morning cortisol, CRP, IL-6, blood pressure, pulse, and PSS/GAD-7. "Alternative activities" were brief, low-cost options (mindfulness/breathing, brisk walking, journaling, in-person peer time), with ≥ 2 modalities per participant.

Results The detox + activities group showed the largest improvements across biomarkers, HRV, and psychometric scores; the reduction-only group showed moderate improvements, while the control group exhibited no significant change. Beyond statistical significance, practical significance was observed: psychometric scores shifted across standard clinical categories (e.g., PSS high to moderate; GAD-7 moderate to mild), HRV increases indicated greater overall heart-rate variability and improved autonomic balance, and modest blood-pressure changes reflected favorable autonomic balance. Qualitative themes (e.g., mental reset, regaining academic control, reconnecting offline) converged with these changes.

Conclusion A structured, low-cost 2-week digital detox, particularly when paired with alternative activities, was associated with psychological well-being and key physiological markers in medical students. Limitations include the short intervention duration, two-site sample, reliance on self-report/app analytics for compliance, and absence of blinding. Digital-use hygiene paired with brief offline activities may offer a scalable student well-being strategy, especially in resource-limited settings; multi-site factorial trials with longer follow-up are warranted.

Trial registration Clinical trial number not applicable.

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Keywords Digital detox, Stress markers, Inflammatory markers, Oxidative stress, Mental health, Medical students

Introduction

Digital technology has transformed medical education, improving access to online resources, simulations, and collaborative learning opportunities. Alongside these benefits, patterns of excess, timing, and context of use, late-night scrolling, "always-on" messaging, and multitasking, are increasingly linked with stress, sleep disruption, attentional lapses, and reduced academic efficiency [1, 2]. These risks are amplified for medical students, who face heavy workloads, irregular clinical rotations, and high-stakes examinations that compress recovery time and incentivize prolonged screen exposure [3, 4]. In this context, pragmatic approaches that reduce non-essential screen exposure without disrupting core learning are of practical interest among educators and trainees.

In Pakistan, rapid uptake of smartphones and online platforms has reshaped study habits and social life on university campuses. Internet penetration is substantial (≈ 45.7%; >111 million active users in 2024), underscoring the pervasiveness of digital engagement among learners [5]. At the same time, campus mental-health services are limited and often difficult to access. Awareness of the health impacts of digital overuse remains low, and structured interventions and preventive strategies are scarce [6]. This combination of high exposure to digital media and constrained support creates a clear need for brief, low-cost interventions that can be implemented at scale within medical colleges and aligned with existing student-support policies. Our focus on medical students in Pakistan is therefore both population-specific (high academic stress, distinctive schedules) and context-relevant (resource constraints and high digital engagement).

Empirical work links heavier digital engagement with elevated perceived stress and anxiety, poorer sleep quality, and attentional difficulties [1, 2, 7]. However, much of the literature relies on cross-sectional designs or short programs with limited objective measures, making it difficult to infer mechanisms or real-world relevance [8, 9]. Few trials test structured "digital detox" programs in medical students and even fewer integrate psychometrics with physiological/biochemical endpoints in low- and middle-income settings [8, 10]. Addressing this gap has practical importance for institutions deciding whether to adopt digital-wellbeing initiatives.

Biologically, sustained overuse may engage stress effector systems, hypothalamic–pituitary–adrenal (HPA) axis, and autonomic nervous system (ANS), with downstream inflammatory and oxidative changes [11]. These processes should be observable as alterations in autonomic function (e.g., lower heart-rate variability [HRV], higher resting pulse and blood pressure) and in stress-related biomarkers (e.g., cortisol; inflammatory markers such as C-reactive protein [CRP] and interleukin-6 [IL-6]; oxidative indices such as malondialdehyde [MDA], superoxide dismutase [SOD], and catalase) [12, 13]. When paired with validated instruments, the Perceived Stress Scale (PSS) and Generalized Anxiety Disorder-7 (GAD-7), these measures enable a multidimensional appraisal of strain and recovery [14]. We intentionally focus on these decision-relevant measures to balance mechanistic insight with interpretability for clinicians and educators.

Our objective was to evaluate whether participation in a brief, structured digital-detox program improves perceived stress and anxiety and favorably shifts autonomic and stress-related biochemical markers. The present study evaluates a digital-detox program for medical students in Pakistan, integrating validated psychometric measures with objective indicators of autonomic function and stress biology. Grounded in the stress recovery model, reducing digital load is expected to lower the body's stress burden and facilitate physiological recovery, as reflected in improved autonomic balance (as measured by HRV) and a reduction in perceived stress. Self-regulation theory further predicts that intentional constraints on device use enhance executive control and perceived mastery over academic tasks, leading to downstream improvements in mood and sleep. The hypotheses of this study are as follows:

H1: The intervention group will show greater reductions in perceived stress (ΔPSS) than controls.

H2: The intervention group will show greater improvements in autonomic balance (ΔHRV metrics) than controls.

H3: Qualitative themes (e.g., *mental reset*, *academic control*) will converge with the direction of quantitative effects.

These findings inform curriculum-embedded digital hygiene, such as structured device-light study blocks and mentorship touchpoints, to support well-being and learning in medical education.

Methods

Study design

This was a prospective interventional, three-group randomized-controlled study designed to assess the effects of a structured 2-week digital detox program on mental health, stress-related biochemical markers, and physiological responses among medical students. A mixed-methods approach was employed, integrating quantitative psychometric evaluations, biochemical analyses, and physiological assessments with qualitative

insights from participant experiences to comprehensively examine the impact of digital detox.

Blinding

Given the behavioral nature of the intervention, participants could not be blinded. However, laboratory technicians and data analysts were masked to group assignment to reduce assessment bias.

Mixed-methods integration

A convergent parallel design was used with equal weighting: quantitative outcomes (psychometrics, HRV, BP, pulse, biomarkers) and focus-group themes were analyzed separately and then integrated via (i) a joint display (Supplementary Table S1) mapping domains (stress, sleep, attention, social connection) to group-level effects and exemplar quotes, and (ii) weaving of convergent/divergent findings in the Discussion. The qualitative strand comprised four post-intervention focus groups (8–10 participants) purposively sampled across campus, year, and gender; sessions were audio-recorded/transcribed verbatim and coded using reflexive thematic analysis by two independent coders with peer-debriefing and an audit trail. We also used code–outcome matrixing to link participant-level changes (e.g., ΔPSS, ΔHRV) with themes, clarifying mechanisms for significant changes and contextualizing modest/null effects (e.g., bedtime-routine barriers).

Study Population

The study was conducted at two medical colleges in Pakistan: Quaid-e-Azam Medical College, Bahawalpur (Public), and Shahida Islam Medical Complex, Lodhran (Private) in September 2024. We purposively selected one public and one private college to capture variation in institutional context while maintaining feasibility for laboratory processing and oversight.

Sampling technique We used stratified random sampling by campus (public vs. private) × academic year (Y1–Y5) × gender. Within each stratum, we created an ordered list from the registrar's roster, applied a random start, and sampled with equal allocation targets per stratum. If a sampled student declined, we replaced them with the next random number in the same stratum to preserve representativeness. Randomization to arms used permuted blocks (sizes 6 and 9) within strata to maintain balance. Allocation was concealed using sequentially numbered, opaque, sealed envelopes prepared by an independent statistician within each stratum.

Sample Size

The sample size was calculated using a statistical formula for interventional studies. Assuming a medium effect size (Cohen's d = 0.5), with 80% power and a significance level of 0.05, the required sample size was 240 participants, randomly assigned in a 1:1:1 ratio to one of three groups:

Group A (Digital Detox + Alternative Activities, n = 80)

Participants underwent a 2-week structured digital detox program, incorporating screen-time restrictions alongside engagement in alternative activities such as mindfulness, exercise, and social interactions.

Group B (Screen-Time reduction Only, n = 80)

Participants reduced their digital screen exposure to the same extent as Group A but did not engage in any structured alternative activities.

Group C (Control Group, n = 80)

Participants continued their regular screen-time habits without any imposed restrictions.

Inclusion criteria

The study included medical students between the ages of 18 and 25 years. Participants were required to use digital devices, such as smartphones, tablets, or laptops, for at least four hours daily for either academic or recreational purposes. Additionally, only students who were willing to participate and provided written informed consent were included in the study.

Exclusion criteria

Students diagnosed with chronic physical or mental health conditions requiring regular medication were excluded from the study. Additionally, those who attended counseling for mental health issues during the study were not included. Students who refused to adhere to the digital detox protocol were also excluded.

Intervention

In the intervention group, recreational screen use (like social media and gaming) was capped at one hour daily, and academic use at two hours. Non-essential screen time was banned from 9:00 AM to 5:00 PM, except for academic needs, and all screens were avoided one hour before bedtime (9:00 PM to 10:00 PM) to improve sleep and nervous system regulation. Participants engaged in alternative activities like mindfulness, exercise, journaling, and in-person socializing, supported by faculty mentors and peer groups for accountability.

The screen-time reduction group followed the same screen limits but didn't participate in alternative activities, helping assess whether screen reduction alone drove improvements. The control group maintained their usual digital habits without restrictions.

We computed a 0–3 Composite Adherence Score per day: (1) app-logged screen-time within limits, (1) screenshot verification received on the scheduled day,

(1) activity log completed (Group A only). The Composite Adherence Score was adapted from multi-indicator adherence approaches used in eHealth interventions; it is not externally validated as a standalone scale, so we report its distribution and associations with outcomes to support construct validity. Non-adherent days triggered same-day reminders. We pre-specified ≥ 75% adherent days as "compliant". To mitigate self-report bias, we combined device analytics with random screenshot audits; nonetheless, we acknowledge residual reporting error. Operational enforcement included in-person configuration of system screen-time controls (Android Digital Wellbeing/iOS Screen Time) with passcodes held by study staff; participants uploaded daily 20:00 dashboard screenshots, and random audits twice weekly cross-checked uploads against app logs. Non-adherence prompted app reconfiguration and a brief troubleshooting call. The framework of this digital detox program is shown in Fig. 1.

Data Collection

Biochemical parameters were assessed from venous blood (IL-6, CRP, MDA, SOD, CAT), and morning salivary cortisol was collected upon awakening and before breakfast at baseline and post-intervention. Salivary cortisol and IL-6 were quantified by electrochemiluminescence immunoassay (Cobas e411, Roche Diagnostics). Twenty-four–hour urine catecholamines (epinephrine, norepinephrine) were collected in acidified containers (10–15 mL 6 N HCl; target pH < 3), kept at 2–8 °C during collection, aliquoted, stored at − 80 °C, and analyzed by high-performance liquid chromatography with electrochemical detection (HPLC/ECD); results are expressed as μg/day. During the 24-h collection, participants avoided caffeine, bananas, citrus, vanilla-containing foods, and decongestants, and abstained from nicotine and vigorous exercise. C-reactive protein (CRP) was analyzed on a Cobas c303 automated analyzer. Oxidative stress indices were quantified using standard spectrophotometric assays: MDA by TBARS at 532 nm; SOD activity by NBT reduction at 560 nm; and catalase (CAT) activity by H_2O_2 decomposition at 240 nm.

Physiological measurements included heart rate variability (HRV), blood pressure (BP), and pulse rate. To reduce circadian and environmental variability, all clinic-based pre- and post-assessments were scheduled 08:00–10:00, fasting (water allowed), after ≥ 12 h abstinence from vigorous exercise and caffeine, following a 10-minute seated acclimation in a quiet room (22–24 °C). HRV (SDNN) was derived from 5-minute supine recordings with spontaneous breathing using a Polar H10; SDNN was the prespecified HRV endpoint, where higher values indicate greater overall heart rate variability. Home BP was measured each morning per American Heart Association (AHA) guidance using an automated digital sphygmomanometer (two consecutive readings ~ 1 min apart, averaged; appropriate cuff size; arm supported at heart level); the resting pulse rate was taken from the device or, where unavailable, by radial palpation averaged over three consecutive measurements. Psychometric assessments used the Perceived Stress Scale (PSS) and Generalized Anxiety Disorder-7 (GAD-7) at baseline and post-intervention. We also recorded habitual sleep duration, caffeine intake (cups/day), and basic sleep-hygiene practices at both time points. Additionally, qualitative data were collected through four post-intervention focus-group discussions exploring subjective experiences, challenges, benefits, and behavioral changes during the digital-detox program.

Data Analysis

Descriptive statistics (mean ± SD) were used to summarize demographic and clinical data. Normality of continuous variables was assessed using the Shapiro–Wilk test. The test revealed non-normal distribution for all key biochemical and psychometric variables: cortisol ($W = 0.935$, $p < 0.001$), CRP ($W = 0.927$, $p < 0.001$), IL-6 ($W = 0.942$, $p < 0.001$), MDA ($W = 0.911$, $p < 0.001$), SOD ($W = 0.956$, $p = 0.003$), catalase ($W = 0.948$, $p = 0.005$), PSS ($W = 0.962$, $p = 0.004$), and GAD-7 ($W = 0.951$, $p = 0.006$). Based on these findings, non-parametric tests were applied throughout the analysis. Within-group comparisons were conducted using the Wilcoxon signed-rank test, while between-group comparisons across the three study arms were performed using the Kruskal-Wallis test. For multiple comparisons, Bonferroni correction was applied where appropriate to control for type I error. Correlation analyses were carried out using Spearman's rank correlation due to non-normal distributions, and effect sizes: $r = Z/\sqrt{N}$ for Wilcoxon (within-group); ε^2 (epsilon-squared) for Kruskal–Wallis (between-group, 3 arms). Values of $r = 0.1$, 0.3, and 0.5 were interpreted as small, medium, and large effect sizes, respectively. Randomization within strata (campus × year × gender) reduced systematic imbalance. Primary inferences used change-scores with the Wilcoxon/Kruskal–Wallis. As a sensitivity check, we fit rank-based ANCOVA models for post-scores controlling for baseline value, gender, academic year, baseline sleep duration, and caffeine intake. We used rank-based ANCOVA (aligned-rank) because it is robust to non-normality and heteroscedasticity while retaining baseline adjustment for greater precision; exact p-values and effect sizes were reported. Primary outcomes were ΔPSS, ΔGAD-7, and ΔHRV (ΔSDNN). Within-group changes used the Wilcoxon signed-rank; between-group differences in change used Kruskal–Wallis (3 arms). For multiplicity, family-wise error was controlled for primary outcomes using Bonferroni;

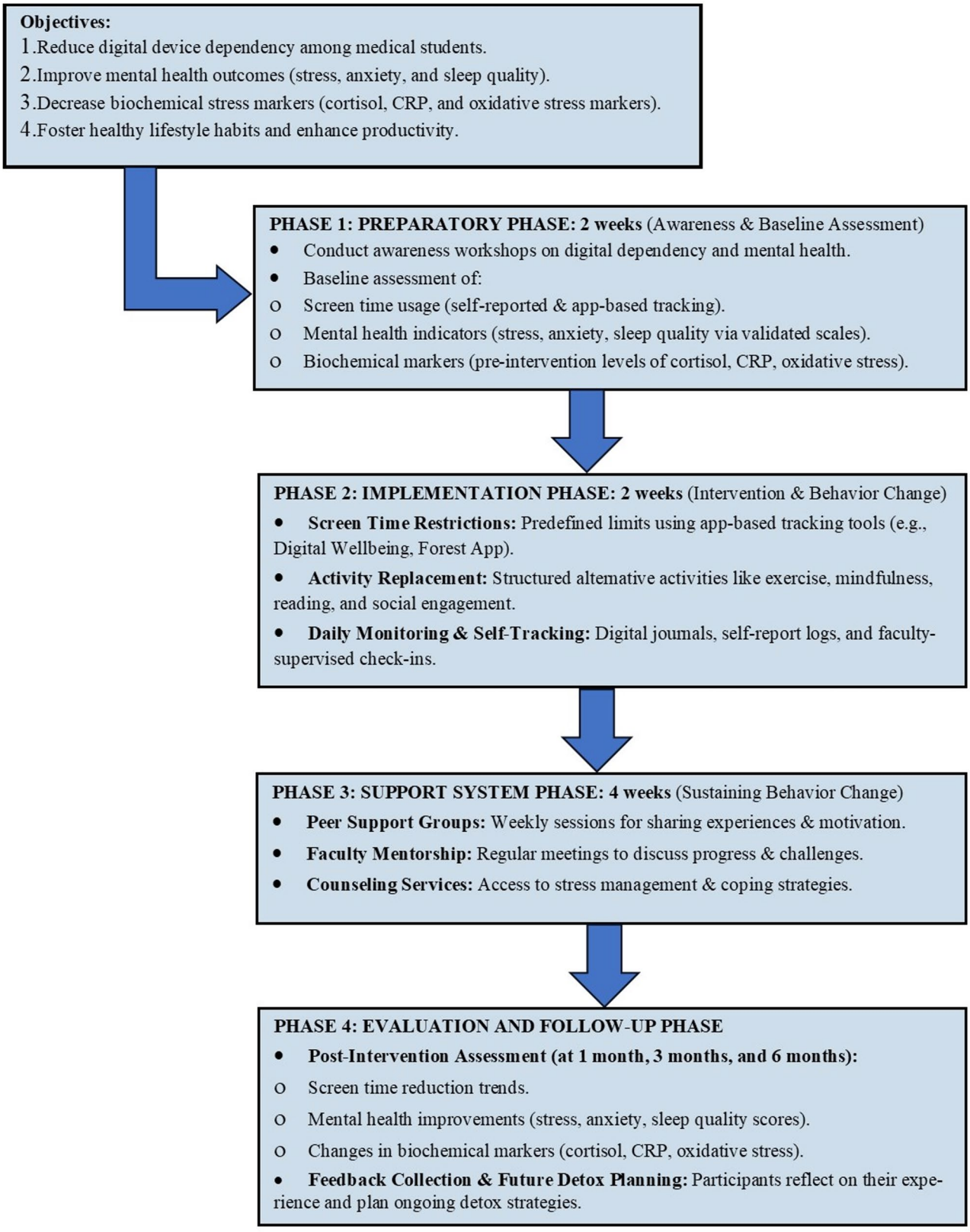


Fig. 1 Framework of Digital Detox Program Phases 3 and 4 depict planned extensions (support and longer-term follow-up) and are not analyzed in the present report. The current analyses are limited to Phases 1 and 2 only

secondary analyses were treated as exploratory with adjusted p-values reported where applicable. For missing data, analyses followed intention-to-treat; for ≤ 1 missing post value, last observation carried forward was used, with complete-case sensitivity analyses yielding similar inferences. For qualitative analysis, two trained coders independently applied the finalized codebook (NVivo). Inter-coder agreement was quantified using Cohen's κ before adjudication; we report overall κ (and range across major codes) in results. Discrepancies were resolved by discussion (third reviewer as needed). Quantitative and qualitative strands were integrated via a joint display and code–outcome matrixing (linking ΔPSS/ΔHRV with themes); saturation was reached when no new codes emerged by the fourth group. All statistical analyses were performed using SPSS version 28, with two-sided significance set at $p < 0.05$.

Ethical considerations

Ethical approval was granted by the Institutional Review Boards of Quaid-e-Azam Medical College and Shahida Islam Medical College. The study was conducted in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Declaration of Helsinki and its later amendments (2013). Participation was voluntary, with no academic penalties for refusal or withdrawal. Faculty involved in the intervention had no academic authority over participants to avoid bias. Participants received clear verbal and written explanations of the study's purpose, procedures, and confidentiality measures. Recruitment notices were distributed via campus email lists and noticeboards by the Student Affairs office; classroom announcements were not used. "Independent facilitators" were staff unaffiliated with course grading or clinical evaluations and were trained to deliver a neutral script emphasizing voluntariness. No course credit or differential compensation was offered. These procedures minimized coercion risk. Device analytics/screenshots and laboratory data were de-identified and stored on encrypted drives with access restricted to the research team; only aggregated data were used for analysis. Participants were advised to avoid substances that could influence biomarkers (e.g., caffeine, nicotine, decongestants) during the pre-assessment abstinence window; adverse events were monitored, and none were reported. The authors used AI-based writing assistance (grammar/style only) to improve clarity. All content, analysis, and conclusions were conceived and verified by the authors. No AI tools were used for data collection, coding, statistical analysis, or the generation of results.

The complete flow of participants from enrollment through allocation, follow-up, and analysis is shown in Fig. 2.

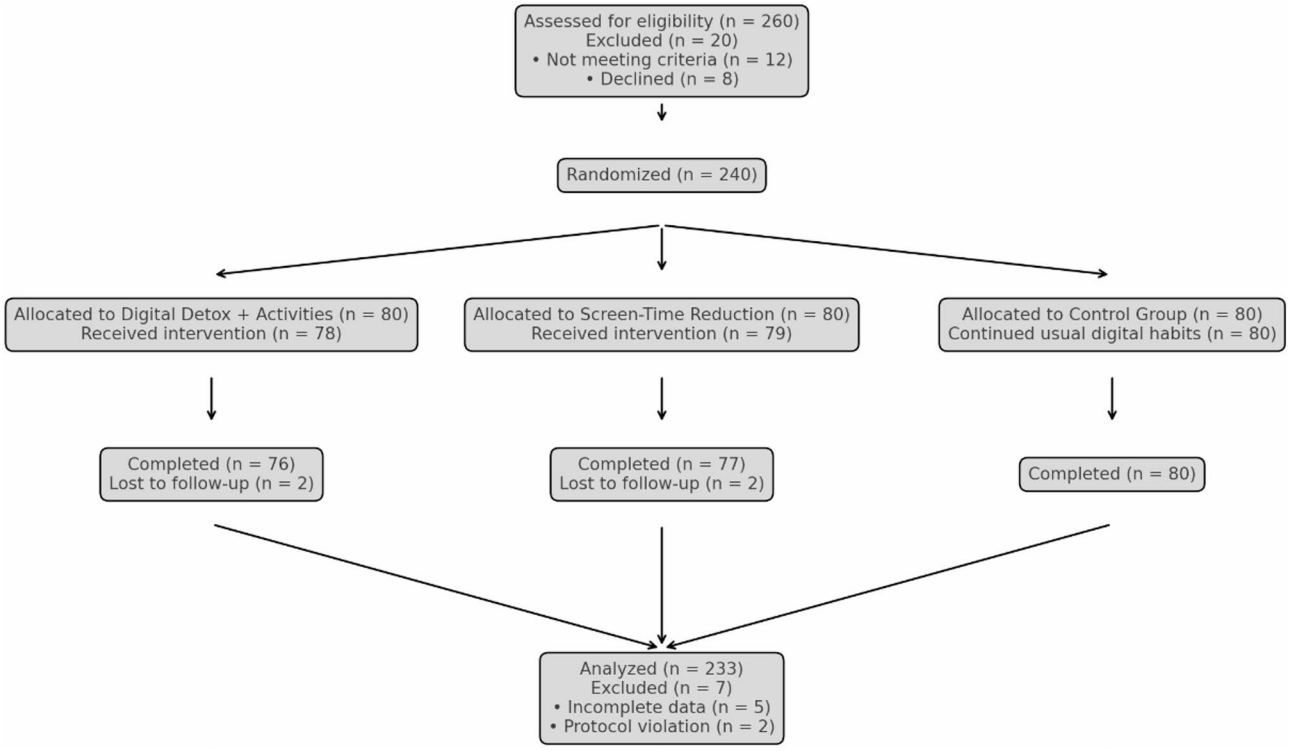


Fig. 2 CONSORT-style flow diagram showing participant recruitment, allocation, follow-up, and inclusion in final analysis

Results

A total of 240 medical students participated in the study, with an equal distribution across the three study groups. The mean age of participants was 22.1 years (SD = 2.4 years), with a female predominance (58%, $n = 139$) and male representation (42%, $n = 101$). Participants were enrolled across five academic years, ensuring balanced representation. The mean daily screen time across all participants at baseline was 6.7 h (SD = 1.5 h). Baseline demographics of study participants are summarized in Table 1.

Retention was 240/240 (100%) through the post-intervention assessment. Composite adherence ≥ 75% was achieved by 90.0% (72/80) in the detox + activities group and 82.5% (66/80) in the reduction-only group (0/80, 0% in controls by design).

Physiological Measurements

The average systolic blood pressure (SBP) for all participants was 121.5 mmHg (SD = 10.2), and the average diastolic blood pressure (DBP) was 77.3 mmHg (SD = 7.9). Using the mean of the last three mornings in week 2 as the post-intervention BP endpoint (baseline = mean of three mornings pre-randomization), participants in the digital detox + alternative activities group showed a highly significant reduction in SBP to 115.6 mmHg (SD = 8.9) and in DBP to 73.9 mmHg (SD = 6.3) post-intervention, compared to the screen-time reduction-only group (SBP: 119.2 mmHg, SD = 9.4; DBP: 75.8 mmHg, SD = 7.1). Between-group differences in ΔSBP and ΔDBP were significant (Kruskal–Wallis $p < 0.01$). The control group showed no significant changes. ΔHRV (ΔSDNN) increased most in the detox + activities group, with smaller gains in the reduction-only group and minimal change in controls (Kruskal–Wallis $p < 0.001$). Heart rate variability (HRV), measured by SDNN, improved in the digital detox + alternative activities group, with an average of 50.1 ms (SD = 8.4), compared to 44.2 ms (SD = 9.0) in the screen-time reduction group and 39.5 ms (SD = 7.7) in the control group. Pulse rate was reduced in the digital detox + alternative activities group to 71.8 bpm (SD = 6.8), in contrast to 76.5 bpm (SD = 7.1) in the screen-time reduction group and 80.2 bpm (SD = 7.4) in the control group. Between-group differences in Δpulse were significant (Kruskal–Wallis $p < 0.001$). HRV increases and small SBP/DBP and pulse reductions are consistent with greater overall heart-rate variability and improved autonomic balance in young adults.

Table 1 Baseline demographics of study participants

Variable	Total ($n = 240$)	Digital Detox + Alternative Activities ($n = 80$)	Screen-Time Reduction Only ($n = 80$)	Control Group ($n = 80$)
Age (years)	22.1 (2.4)	22.3 (2.5)	22.0 (2.3)	21.9 (2.4)
Female (n, %)	139 (58%)	46 (57%)	47 (59%)	46 (58%)
Male (n, %)	101 (42%)	34 (43%)	33 (41%)	34 (42%)
Academic Year 1 (n, %)	48 (20%)	15 (19%)	17 (21%)	16 (20%)
Academic Year 2 (n, %)	53 (22%)	18 (23%)	17 (22%)	18 (21%)
Academic Year 3 (n, %)	43 (18%)	14 (18%)	15 (19%)	14 (17%)
Academic Year 4 (n, %)	48 (20%)	16 (20%)	15 (19%)	17 (21%)
Academic Year 5 (n, %)	48 (20%)	17 (20%)	16 (19%)	15 (21%)
Screen Time (hours)	6.7 (1.5)	6.6 (1.4)	6.7 (1.5)	6.8 (1.6)

Biochemical analysis

Within-group (Wilcoxon) analyses showed salivary cortisol, CRP, and IL-6 decreased significantly in the digital detox + alternative activities group (all $p < 0.001$; $r = 0.70$, 0.61, 0.59, respectively), with moderate improvements in the screen-time reduction group and no meaningful change in controls. Between-group differences in change (Δ) for these markers were significant (Kruskal–Wallis $p < 0.01$; see Table 2). Twenty-four–hour urinary catecholamines (epinephrine, norepinephrine) and MDA also improved significantly within the intervention arms ($p < 0.05$; $r = 0.49$ and 0.41, respectively). Catalase increased modestly only in the detox group ($p = 0.041$; $r = 0.28$), whereas SOD showed no significant change ($p = 0.187$). In sensitivity analyses, rank-ANCOVA confirmed the three-group pattern for cortisol, CRP, and IL-6 (global $p < 0.01$; Bonferroni-corrected $q < 0.05$). A summary of biochemical marker changes across groups is presented in Table 2.

Psychometric assessments Within-group (Wilcoxon) analyses showed Perceived Stress Scale (PSS) and Generalized Anxiety Disorder-7 (GAD-7) scores decreased significantly in the digital detox + alternative activities group (both $p < 0.001$; $r = 0.65$ and 0.61, respectively). The screen-time reduction group showed moderate improvements, whereas the control group exhibited no meaningful change. Between-group differences in change (Δ) were significant (Kruskal–Wallis $p < 0.001$). In sensitivity analyses, rank-ANCOVA confirmed the three-group pattern for PSS and GAD-7 (global $p < 0.01$; Bonferroni-corrected $q < 0.05$ for co-primary outcomes). Details of pre- and post-intervention psychometric scores across all groups are summarized in Table 3.

Changes corresponded to category shifts (e.g., PSS high to moderate; GAD-7 moderate to mild), indicating practical relevance beyond statistical significance.

Digital usage Data from the app tracking software revealed significant differences in screen time between the intervention and control groups. Participants in the intervention group lowered their daily screen usage

Table 2 Changes in biochemical markers Pre- and Post-Intervention

Variable	Digital Detox + Activities (Pre → Post)	Screen-Time Reduction Only (Pre → Post)	Control Group (Pre → Post)	*p*-value
Morning Cortisol (nmol/L)	15.7 (4.4) → 12.8 (3.9)	15.9 (4.5) → 14.2 (4.2)	15.1 (4.8) → 15.0 (4.7)	< 0.001
24 h Urinary Epinephrine (µg/day)	35.2 (9.4) → 30.1 (8.1)	34.7 (9.3) → 33.9 (9.0)	34.5 (9.5) → 34.4 (9.6)	< 0.05
24 h Urinary Norepinephrine (µg/day)	110.5 (24.6) → 102.8 (22.8)	109.9 (22.6) → 104.4 (22.1)	110.3 (23.8) → 109.8 (22.8)	< 0.05
MDA (nmol/mL)	2.9 (0.9) → 2.1 (0.5)	2.8 (0.8) → 2.6 (0.6)	2.9 (0.8) → 2.9 (0.7)	< 0.01
SOD (U/mg protein)	50.7 (12.6) → 51.1 (11.5)	50.3 (12.4) → 50.3 (10.9)	50.1 (12.3) → 49.8 (9.8)	0.187
Catalase (U/mg protein)	30.3 (7.8) → 32.3 (8.0)	30.5 (7.6) → 31.2 (7.1)	30.2 (7.7) → 30.0 (7.3)	0.041
CRP (mg/L)	2.9 (0.8) → 1.7 (0.6)	2.8 (0.9) → 2.3 (0.7)	2.7 (0.9) → 2.7 (0.8)	< 0.001
IL-6 (pg/mL)	4.3 (1.2) → 2.6 (1.0)	4.1 (1.3) → 2.8 (1.1)	4.2 (1.3) → 4.2 (1.2)	< 0.001

*Data are presented as Mean (SD). Shapiro-Wilk test indicated non-normal distribution for all biochemical variables ($p < 0.01$). Between-group comparisons were performed using the Kruskal-Wallis test with Bonferroni correction. Within-group: Wilcoxon (effect size ***r***). Between-group: Kruskal–Wallis (effect size ε^2)

Table 3 Changes in psychometric scores Pre- and Post-Intervention

Variable	Digital Detox + Activities (Pre → Post)	Screen-Time Reduction Only (Pre → Post)	Control Group (Pre → Post)	*p*-value
PSS Score	26.2 (4.2) → 17.8 (3.5)	26.0 (4.3) → 21.6 (4.2)	25.9 (4.5) → 25.7 (4.5)	< 0.001
GAD-7 Score	12.4 (3.7) → 7.1 (3.0)	12.2 (3.6) → 10.3 (3.4)	12.1 (3.7) → 12.0 (3.7)	< 0.001

*Data are presented as Mean (SD). PSS and GAD-7 scores were non-normally distributed based on the Shapiro-Wilk test ($p < 0.01$). Kruskal-Wallis test with Bonferroni correction was used for between-group comparisons; Within-group: Wilcoxon (effect size ***r***). Between-group: Kruskal–Wallis (effect size ε^2)

by 53% from 6.6 h to 3.1 h. The control group reported minimal changes as they reduced their screen time from 6.8 to 6.7 h only. This reduction aligns with the observed improvements in mental health and stress markers, reinforcing the efficacy of the digital detox program. Tracking data confirmed adherence to screen time limits in the intervention group.

Table 4 Correlation analysis between screen Time, mental health Scores, and biochemical stress markers

Variable	Baseline Screen Time (*r*)	Post-Intervention Screen Time (*r*)	*p*-value
Perceived Stress Scale (PSS)	+0.68	−0.61	< 0.001
Generalized Anxiety Disorder-7 (GAD-7)	+0.63	−0.58	< 0.001
Cortisol	+0.72	−0.64	< 0.001
Catecholamines	+0.50	−0.41	0.01
C-reactive Protein (CRP)	+0.65	−0.57	< 0.001
Interleukin-6 (IL-6)	+0.60	−0.52	< 0.01
Malondialdehyde (MDA)	+0.52	−0.44	< 0.01
Superoxide Dismutase (SOD)	−0.10	+0.12	0.42
Catalase	−0.18	+0.21	0.08
Heart Rate Variability (HRV)	−0.58	+0.62	< 0.001
Systolic Blood Pressure (SBP)	+0.53	−0.50	< 0.001
Diastolic Blood Pressure (DBP)	+0.49	−0.46	< 0.001
Pulse Rate	+0.56	−0.51	< 0.001

*Spearman's rank correlation coefficients are reported due to non-normality of all variables as confirmed by Shapiro-Wilk test ($p < 0.01$)

Correlation analysis Baseline screen time correlated positively with cortisol/CRP/PSS and negatively with HRV; post-intervention screen time showed the opposite pattern. These associations are directional and non-causal and complement the group comparisons. Table 4 shows the correlation analysis between screen time, mental health scores, and biochemical stress markers.

Not all biochemical markers changed uniformly (e.g., SOD: ns), indicating heterogeneous responsiveness over 2 weeks; interpretation is provided in the Discussion.

Mixed-methods integration A joint display (Supplementary Table S1) maps each domain (stress, sleep, attention, social connection) to group-level effects (Δ post–pre; A vs. B vs. C) with exemplar quotes, showing convergence for stress/anxiety and autonomic indices and partial convergence for sleep.

Exploratory component contrast (A vs. B) Comparing Detox + Activities (A) to Reduction-only (B) showed larger improvements in PSS, GAD-7, HRV (SDNN), and cortisol for A (Mann–Whitney tests; exploratory). Estimates and effect sizes are in Supplementary Table S2. This contrast suggests the added benefit of alternative activities, while the trial did not isolate individual activity types or content categories.

Qualitative insights

The post-program group meetings showed that the participants in the treatment group experienced both improved personal habits and mental well-being. Our thematic analysis produced three main themes along with their supporting subthemes that represent what

participants learned from their digital detox experience. The findings show participants gained better attention skills with improved sleep and felt less anxious, as confirmed by what these individuals shared about their experiences. Table 5 outlines the qualitative themes and subthemes derived from participant experiences following the intervention.

Supplementary Table S1 aligns effect sizes (r) for primary outcomes with representative themes/quotes (e.g., Mental reset mapping onto reduced PSS and improved HRV). This triangulation showed strong convergence for stress/anxiety and autonomic indices; sleep-related outcomes showed partial convergence, consistent with reported caffeine use and bedtime habits. See Supplementary Table S1 for the full joint display and integrated interpretation.

Table 5 Themes and subthemes identified

Main Theme	Subthemes	Illustrative Quote
Unplugging the Habit	- Realizing excessive screen time - Emotional discomfort during restriction - Confronting dependency	"I didn't realize how hooked I was until I had to actually stop using my phone. I felt agitated for the first few days." (Participant 43, Year 3) "My screen time was shocking. I had no idea it was over 9 hours." (Participant 58, Year 2)
Mental Reset	- Improved clarity and focus - Emotional stability - Reduced anxiety and overstimulation	"By the fourth day, I felt mentally lighter… like a fog had lifted." (Participant 112, Year 4) "I wasn't as anxious as I used to be. It's like my brain stopped buzzing." (Participant 165, Year 4)
Regaining Academic Control	- Fewer distractions during study - Improved concentration span - Better academic scheduling	"I was able to study for 2 hours straight without checking my phone. That's a big deal for me." (Participant 89, Year 3) "I finally finished a topic without having to re-read it five times." (Participant 133, Year 1)
Reclaiming Rest	- Increased awareness of bedtime use - Earlier sleep initiation - Enhanced sleep quality	"I started sleeping at 10:30 pm without scrolling for hours. I woke up fresher." (Participant 176, Year 2) "I had dreams again—like real ones. Probably because I wasn't watching anything before bed." (Participant 121, Year 3)
Reconnecting Offline	- Playing board games and group discussions - Rebuilding face-to-face communication - Improved family time	"We played Ludo after dinner for the first time in years. It made our evenings feel more real." (Participant 203, Year 4) "Talking to my sister instead of texting her from the next room—felt weird at first, but good." (Participant 97, Year 2)

Discussion

The findings of this study indicate that a 2-week digital detox significantly reduced psychological stress and anxiety, accompanied by notable decreases in physiological stress markers. The most pronounced biochemical changes were reductions in cortisol (↓18%), CRP (↓41%), IL-6 (↓40%), catecholamines—epinephrine (↓14%) and norepinephrine (↓7%), and MDA (↓28%), with minor changes in catalase (↑7%) and no significant alterations in SOD (~0%). HRV improved [post 50.1 ms; vs. 44.2/39.5 in comparators], while SBP [post 115.6 mmHg], DBP [post 73.9 mmHg], and pulse [post 71.8 bpm] decreased, reinforcing the role of autonomic regulation in stress mitigation. However, the magnitude of improvement varied among markers, likely reflecting differences in physiological adaptation to stress recovery. Beyond statistical significance, psychometric changes corresponded to category shifts (e.g., PSS high to moderate; GAD-7 moderate to mild), supporting practical/clinical relevance. Given the 2-week duration, effects should be interpreted as short-term adaptations.

Quantitative reductions in PSS/GAD-7 and increases in HRV converged with themes of "mental reset" and "regaining academic control," while the modest sleep changes diverged from some narratives, consistent with reported caffeine/bedtime habits. This integration strengthens inference about the mechanism (reduced arousal/overstimulation) while flagging modifiable barriers to sleep improvement. The mixed-methods joint display summarizing these linkages is provided in Supplementary Table S1. In educational practice, mental reset supports embedding short device-light study blocks and a brief bedtime digital-hygiene routine, while regaining academic control aligns with scheduled faculty mentorship touchpoints and weekly peer support to scaffold self-regulation within the curriculum.

The substantial reduction in cortisol aligns with previous studies demonstrating that behavioral stress-reduction interventions rapidly modulate HPA axis activity [15, 16]. Similarly, the decrease in CRP and IL-6 corroborates prior findings suggesting that digital overstimulation contributes to chronic low-grade inflammation, which can be reversed by limiting screen exposure [12, 17, 18]. The moderate decrease in catecholamines is consistent with the literature, indicating that sympathetic nervous system adaptations occur more gradually compared to cortisol-driven responses [19–21]. However, some studies report minimal autonomic changes following behavioral interventions, suggesting that additional strategies, such as structured relaxation techniques, may be required for full autonomic normalization [22]. The observed pattern is consistent with differential time courses—HPA-axis indices may change within days, whereas inflammatory and autonomic markers often require longer or repeated exposures to consolidate.

The observed decline in MDA suggests a reduction in oxidative stress following digital detox, supporting existing evidence that stress-induced ROS generation contributes to lipid peroxidation [12, 13, 21]. While catalase increased slightly, the lack of a significant change in SOD suggests that endogenous antioxidant defenses may require longer adaptation periods. These findings extend prior research indicating that short-term digital detox primarily mitigates oxidative damage rather than enhancing antioxidant enzyme activity [13, 21]. In contrast, some studies depict a significant improvement in SOD levels after stress relief [23]. Our short intervention window likely limited the capacity to detect enzyme-level adaptations that may emerge over longer follow-up.

Consistent with previous studies, the intervention led to improvements in perceived stress and anxiety, reinforcing the link between biochemical and psychological well-being [14]. Similar to our study, a review states that participants engaging in alternative activities alongside digital detox exhibited the most substantial benefits, aligning with evidence that behavioral engagement enhances stress resilience beyond mere screen-time reduction [24, 25]. However, improvements in sleep quality were modest despite nighttime screen restriction; previous literature has shown a marked improvement in sleep quality [26]. This suggests that additional lifestyle factors, such as caffeine intake and sleep hygiene, influence sleep regulation. The limited sleep gains likely reflect unaddressed factors (caffeine, irregular schedules, light exposure). Incorporating brief sleep-hygiene modules could potentiate detox effects [27]. The qualitative results confirmed the statistical decreases in stress and anxiety. The participants experienced better mental clarity and increased awareness about their screen usage after the detox, which matched the positive changes in PSS and HRV results. The agreement between these different data points confirms that behavioral modifications led to better physiological results. Our exploratory A-vs-B contrast (Detox + Activities vs. Reduction-only) also suggests an added benefit of alternative activities; estimates are reported in Supplementary Table S2 and should be interpreted cautiously as exploratory.

While our design contrasted "detox-only" vs. "detox + activities," it did not quantify the contribution of *individual* activity types or content categories (social media, gaming, educational). Future work should use a multifactorial or SMART design to partition the effects of time restriction, content type, and specific alternative activities. A factorial design would also allow formal testing of interactions (e.g., time limits × activity type) to isolate active components.

Strengths include an objective, multimodal assessment anchored to a prespecified model, which combines validated psychometrics with biochemical stress markers (cortisol, CRP, IL-6, catecholamines, oxidative indices) and physiological measures (HRV, blood pressure, pulse). The three-group design (detox + alternative activities, screen-time reduction only, control) allows an initial separation of the added value of behavioral engagement beyond time restriction. Key limitations are 2-week duration without follow-up (durability unknown), reliance on self-report/device analytics rather than wearables, two medical colleges within one province (generalizability), voluntary recruitment (selection bias), residual circadian/environmental variability despite standardized assessments, and lack of participant blinding (with laboratory technicians and data analysts masked; expectancy/placebo effects still possible). This directly lead to priorities for next studies: extended follow-ups to test durability; incorporation of wearable technology for precise, objective tracking; inclusion of more diverse populations and institutional settings; pre-specified examination of gender differences in stress responses; evaluation of brief, structured sleep interventions alongside digital detox; and pre-registered mediation (e.g., HRV) and moderation analyses (e.g., baseline screen time, caffeine use) to clarify mechanisms and for whom the intervention confers benefit. Future trials should also consider sham or attention-control conditions and blinding where feasible.

Conclusion

This study provides evidence that a 2-week structured digital detox intervention was associated with reduced stress, anxiety, and physiological stress markers, particularly when combined with alternative activities. The observed biochemical improvements in oxidative stress, inflammation, and autonomic balance highlight the physiological benefits of reducing digital overstimulation. While digital detox effectively mitigates acute stress, future research should explore its long-term sustainability and broader applicability across diverse populations. Given the short duration, two-site sampling, and non-blinded participants, these findings should be viewed as preliminary but promising; multi-site, longer-duration, factorial trials are warranted to isolate component effects and assess durability.

Supplementary Information

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Supplementary Material 1.

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Authors' contributions
S.F. conceptualized the study and provided overall supervision. S.R. performed data collection, analysis, and figure/table preparation. S.B. conducted the literature review and contributed to referencing and psychometric assessment. M.F.A. proofread the manuscript and gave final approval. M.I. supported data collection and statistical analysis. All authors reviewed and approved the final manuscript.

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Data availability
The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
The study was approved by the Institutional Review Boards of Quaid-e-Azam Medical College, Bahawalpur, and Shahida Islam Medical College, Lodhran. All participants provided written informed consent before enrollment. Participation was voluntary, and participants could withdraw at any time without penalty. The study was conducted in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Declaration of Helsinki and its later amendments (2013).

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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